Earthquake Damage Predictor

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Problem Statement

Goal is to predict **damage levels to buildings** .

Earthquakes can cause varying levels of structural damage depending on building characteristics and location.

The dataset contains **260,000+ buildings with with 38 features** , including structural, geographical, and usage information

This problem has real-world importance in helping governments and orgonizations prioritize **reconstruction and improve building safety standards** .

We make recommendations based on findings to improve building safety

Challenges

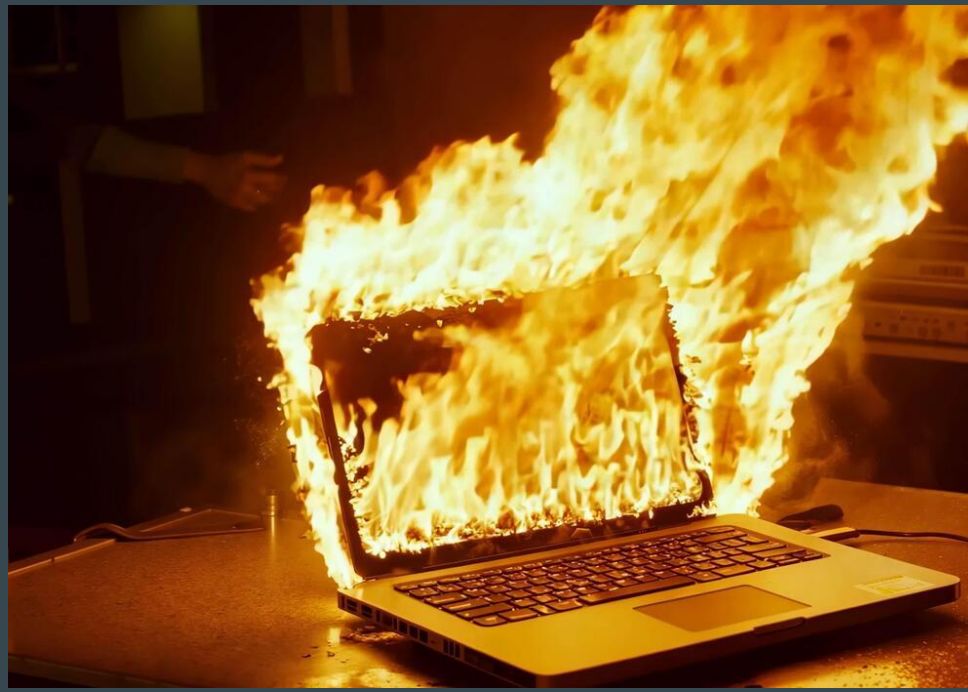
Large, Imbalanced dataset.

High level of noise.

Large number of instances with same features but different outcomes.

Most relevant features difficult to encode (high cardinality)

Other Contestants have had years to train and submit models.



Data Exploration

Entries: 260601 - Building IDs

Variables:38 (numerical and categorical)

Label: Damage category [1-3]

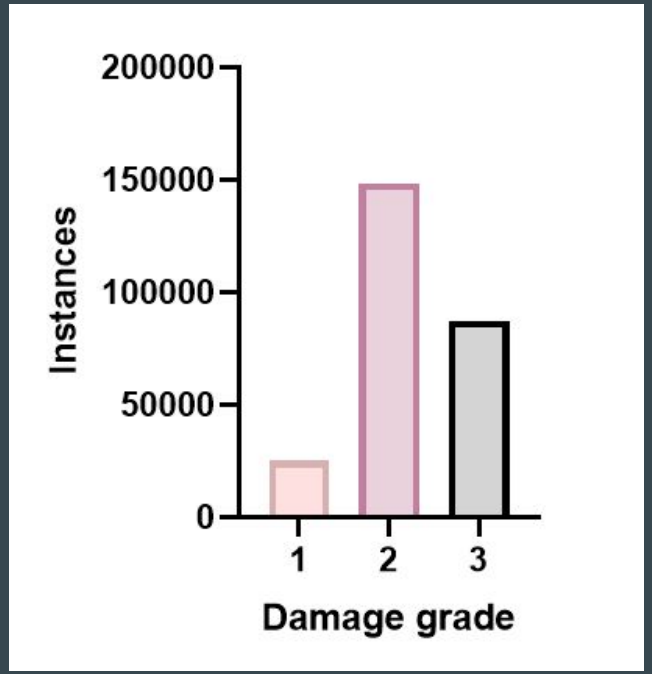
Numerical variables (8)

- Geo ID levels (?) :geographic region in which building exists
- Age of the building in years
- Number of floors
- Normalized area and height of the building footprint
- Number of families

Categorical/Binary variables (30)

- Surface condition
- Foundation, roof and ground type
- Position and plan configuration
- Superstructure composition (e.g. brick, stone, bamboo, etc,)
- Type of secondary use, if any

Damage classes are imbalanced



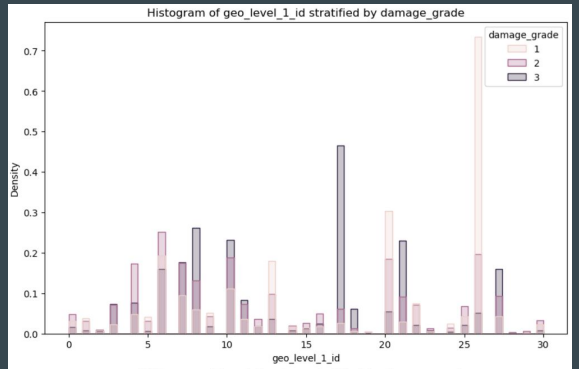
Stratification during Test/Training split: sets maintain the same class proportions as the full dataset

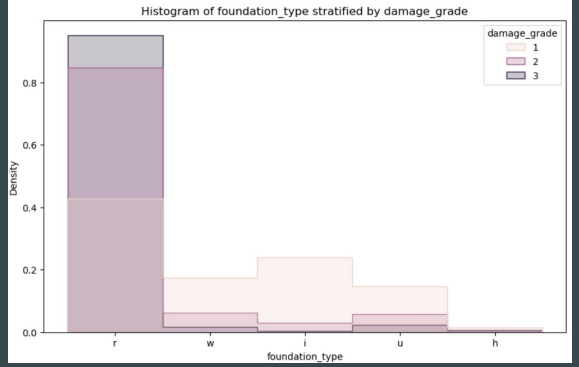
Considering **Balancing** of training set to help the models learn minority classes better.

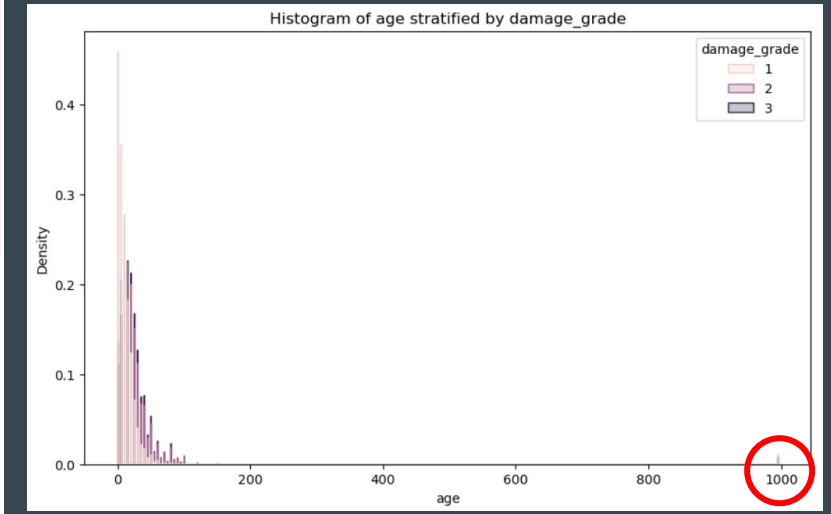
F1 Score for imbalanced data evaluation

Feature distributions by damage class

Histograms with normalized densities for numerical variables and Barplots with normalized proportions for categorical



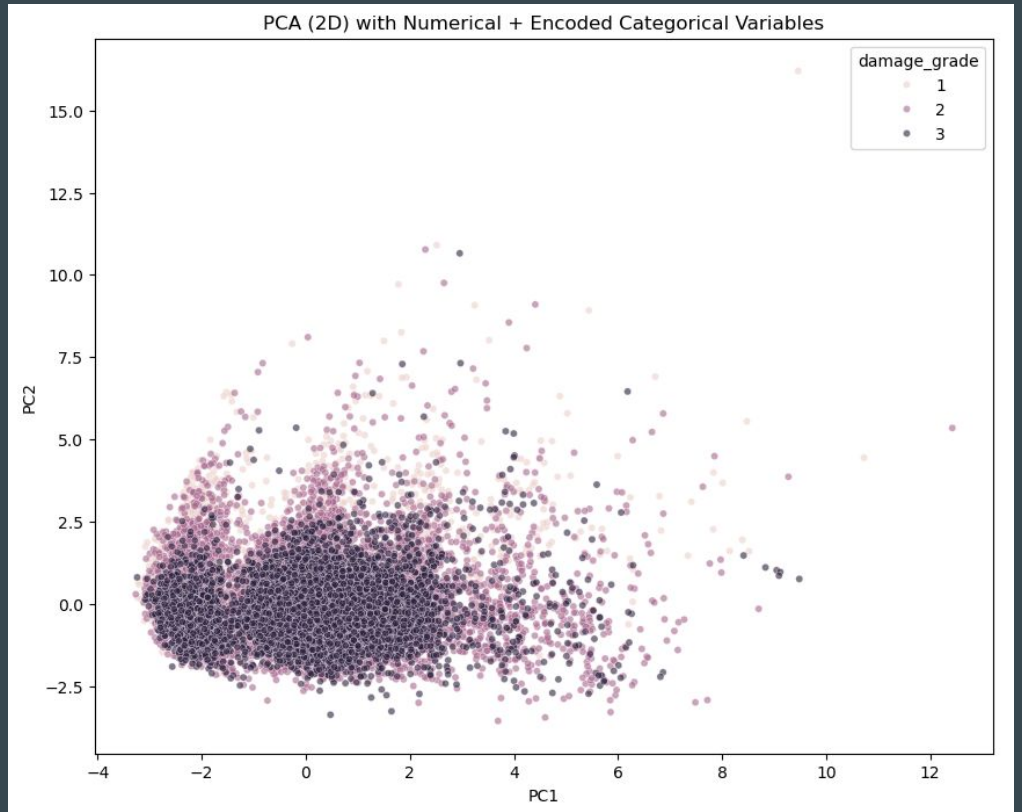




Age = 995

Dimensionality reduction plot (2D PCA plots)

Understanding if damage classes cluster in the encoded feature space



Methodology For overcoming challenges

Test Many approaches

Overfit, then reduce overfitting through regularization, PCA, etc

Hyperparameter tuning

While Gridsearch is searching, research other methodologies

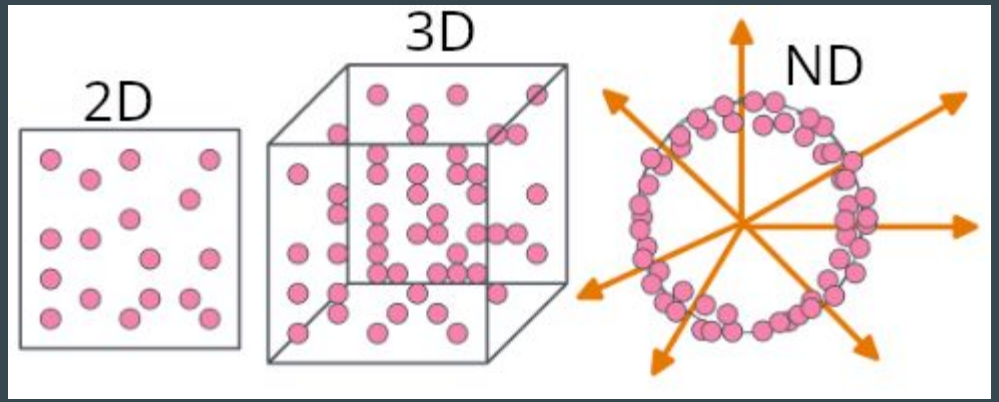
Engineer new features: create features from existing data aiming to improve Performance

Early approaches and the Curse of dimensionality:

KnearestNeighbors Performs worse as dimensions increase.

Support Vector Machines considered, but not ideal for large datasets.

Neural Networks attempted but long training times and inexperience limited effectiveness.



We switched to Tree based classifiers and ensemble methods to model the high dimensional, non linear data.

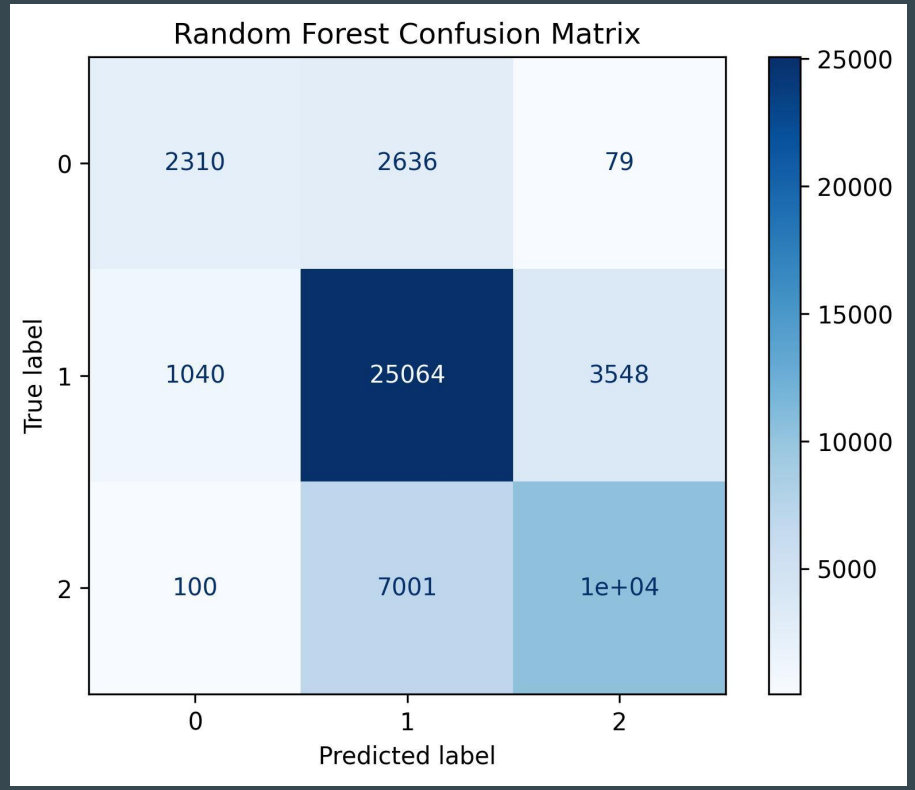
Tree Ensemble methods

Random Forest ensemble methods generalise well on noise, and the models gave us stable performance across splits.

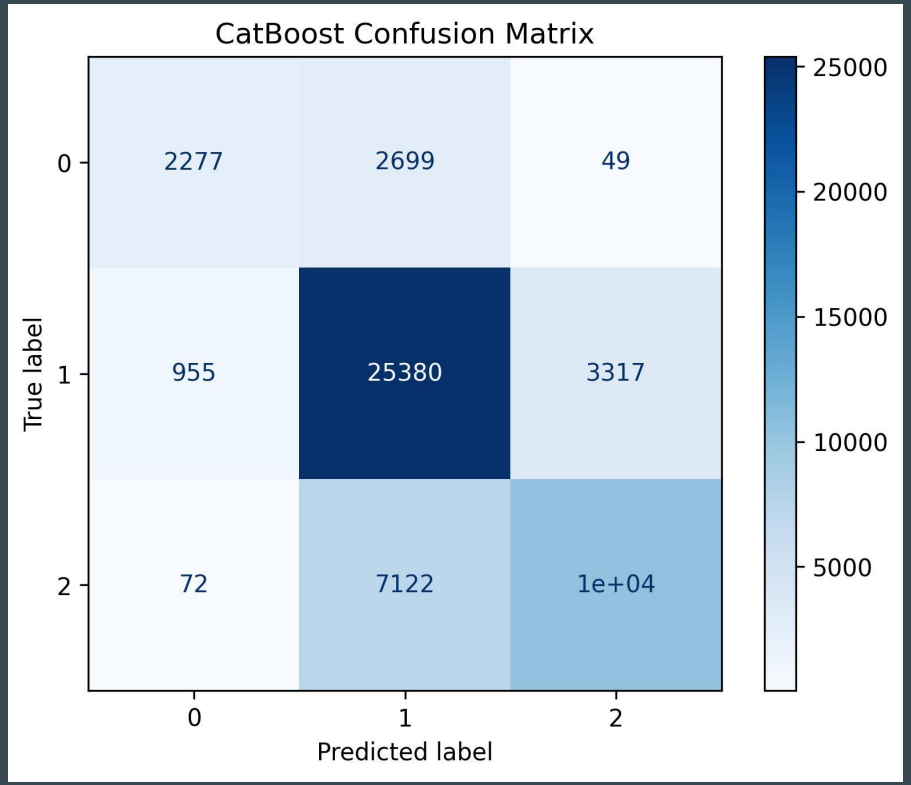
Train/test validation F1: 0.724

Cross-validation F1 (cv=5): 0.724

<u>Tuning:</u>
Increasing the number of trees (from 100 to 300) improved performance and adjusting the minimum samples per split (from 2 to 5) reduced overfitting.



Boosted Trees (Catboost, Histboost)



CatBoost: created to better handle categorical features. In our case, it reduced extreme misclassifications, capturing subtler differences between damage levels. However, it was more sensitive to data splits than Random Forest.

Histogram-base Gradient Boosted trees trade slight variance and interpretability for faster training speed than standard GBT.

Our Histogram Boosted Trees attained the highest competition score at 0.7405

L2 Regularization allowed HistBoost to overcome overfitting

Results

Our solution places us 1146th out of 8722 contestants (just outside top 13%)

#1 Contestant's solution is better than ours by 2% (0.7558 vs our 0.7405 F1_score). Median score is 0.7348

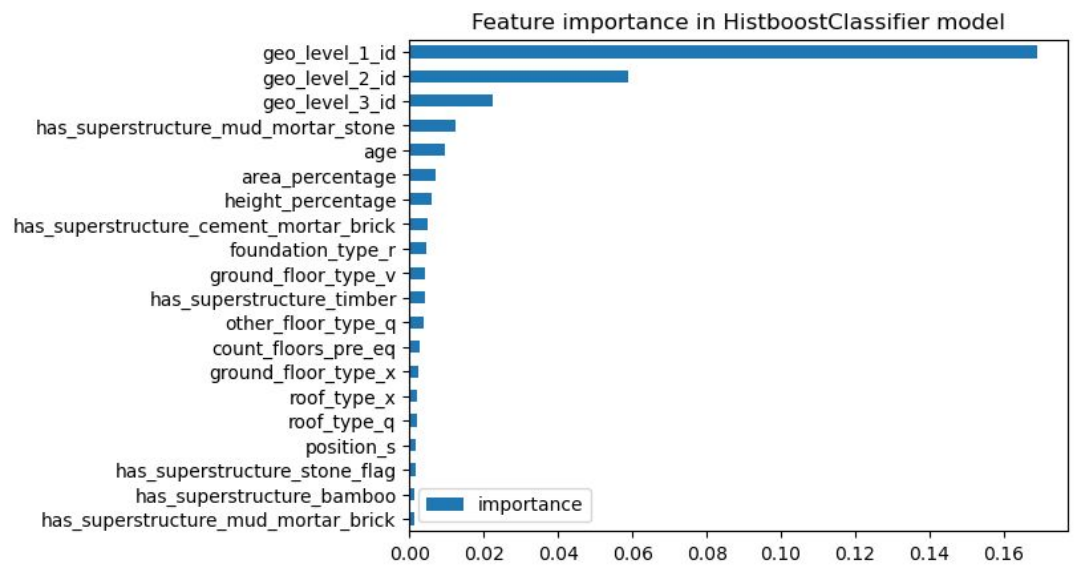
Top contestants have submitted many attempts (some hundreds) compared to 4 for us.

1 year left 8,722 joined

Leaderboard

Rank	Participant	Best public Micro F1	Shared work
#1	inoddy 6y 3mo ago · 64 submissions	0.7558	
#2	seismicnz 6y 3mo ago · 27 submissions	0.7553	
#3	NotHuhu 3y 4mo ago · 18 submissions	0.7542	
#4	Tester22 1y 10mo ago · 270 submissions	0.7542	
#1144	Noel 3y 5mo ago · 1 submission	0.7405	
#1145	SierszulskiM 6y 3mo ago · 7 submissions	0.7405	
#1146	evan2222 3d 23h ago · 4 submissions	0.7405	Share your work!
#1147	SilverWerewolf1024 4y 8mo ago · 2 submissions	0.7404	
#1148	lolzilla 3y 11mo ago · 3 submissions	0.7404	
#1149	CSCI567_id53 2y 11mo ago · 4 submissions	0.7404	
#1150	anu1991 5y 5mo ago · 4 submissions	0.7403	

Interpreting the results: Part 1



Feature importance similar to Catboosted Trees.

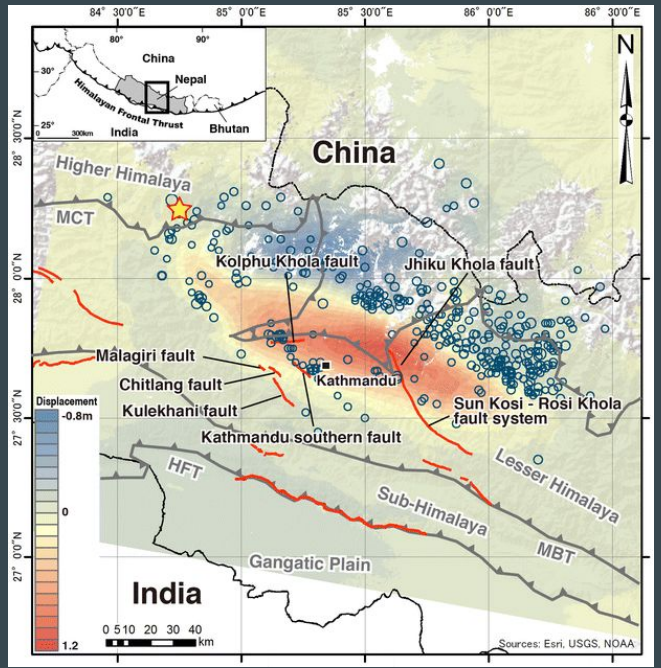
Geographical location is the most important feature, often requiring several tree nodes.

superstructure type, building age of a building, the ratio of height to width, and the age of the building.

Least important categories were building use, legal status.

Interpreting the results: Part 2

We combine feature importance with previous data exploration to make the following suggestions to the Nepales Government:



- Avoid or raise building standards in select geography (fault lines, remote areas, etc).
- Reconstruct at-risk homes out of earthquake resistant materials (Reinforced Concrete, Timber).
- Avoid Mud Mortar based walls.
- Inspect older buildings, regardless of current use.

Other sources generally agree, and also recommend increased response training.

Nepal is a developing country, and so implementing all recommendations at once is unfeasible.

Lessons learned

The most important features may be unexpected

Worry about underfitting first, then overfitting

Not all Feature Augmentation is useful

The curse of dimensionality can make some approaches unfeasible

Reducing dimensionality can fail if key information is degraded

Sources and documentation

DrivenData. (2019). *Richter's Predictor: Modeling Earthquake Damage*. Retrieved [Month Day Year] from https://www.drivendata.org/competitions/57/nepal-earthquake.

Use of AI:

AI was used to answer questions about syntax, assist in troubleshooting errors, and to recall answers to basic questions (eg: ‘how is f1 score calculated?’)